

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Computer Networks

COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Naga Jaswanth B (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Naga Jaswanth B

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = 150 MB = 150,000,000 bytes
- Segment size = 1500 bytes
- Data Link Layer overhead = 40 bytes per frame
- Bandwidth = 50 Mbps

Calculation

Step 1: Calculate the number of segments

Number of segments = File size ÷ Segment size

= 150,000,000 ÷ 1500

= **100,000 segments**

Since one segment is placed into one frame,

Number of frames = **100,000 frames**

Step 2: Calculate the total Data Link Layer overhead

Total overhead = Number of frames × Overhead per frame

$$= 100,000 \times 40$$

$$= \mathbf{4,000,000 \text{ bytes}}$$

$$= \mathbf{4 \text{ MB}}$$

Step 3: Calculate the total data transmitted

Total data transmitted

$$= \text{Original file size} + \text{Overhead}$$

$$= 150 \text{ MB} + 4 \text{ MB}$$

$$= \mathbf{154 \text{ MB}}$$

Step 4: Calculate the transmission time

Convert MB into Megabits

$$154 \times 8$$

$$= \mathbf{1232 \text{ Megabits}}$$

Transmission time

$$= \text{Total bits} \div \text{Bandwidth}$$

$$= 1232 \div 50$$

$$= \mathbf{24.64 \text{ seconds}}$$

Answer

- Number of segments = **100,000**
- Number of frames = **100,000**
- Total overhead = **4 MB**
- Transmission time = **24.64 seconds**

Explanation

The **Transport Layer** divides the file into smaller segments and ensures reliable delivery using sequence numbers, acknowledgements, retransmissions, and flow control. The **Data Link Layer** encapsulates each segment into frames by adding headers and trailers, performs error detection using CRC, and ensures reliable communication between directly connected devices. Together, these layers provide accurate, efficient, and reliable data transmission.

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
- Total frames = 48
- Frame size = 1024 bytes

Calculation

Step 1: Calculate the number of transmission windows

Number of windows

= Total frames ÷ Window size

= $48 \div 6$

= **8 windows**

Step 2: Calculate retransmissions

One frame is lost in every window.

Total retransmissions

= 8×1

= **8 frames**

Answer

- Number of transmission windows = **8**
- Total retransmissions = **8 frames**

Explanation

Flow control regulates the rate of data transmission between the sender and receiver. It prevents receiver buffer overflow, minimizes packet loss, avoids unnecessary retransmissions, and improves communication efficiency. The sliding window protocol allows multiple frames to be transmitted before receiving acknowledgements, thereby increasing network throughput.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead

introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

Calculation

Maximum payload in one fragment

= MTU – Header

= 1500 – 20

= **1480 bytes**

Payload to be fragmented

= 12,000 – 20

= **11,980 bytes**

Number of fragments

= 11,980 ÷ 1480

= 8.09

Therefore,

= 9 fragments

Header overhead

= Number of fragments $\times$ Header size

= 9×20

= 180 bytes

Answer

- Number of fragments = **9**
- Total header overhead = **180 bytes**

Explanation

The **Network Layer** performs fragmentation whenever a packet exceeds the MTU of the network. Each fragment contains its own IP header with identification, fragment offset, and control information. At the destination, these fragments are reassembled correctly to reconstruct the original packet before it is delivered to the upper layers.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
- Total frames = 48

Calculation

Step 1: Number of windows

$$48 \div 6$$

$$= \mathbf{8 \text{ windows}}$$

Step 2: Retransmissions

One frame is lost in every window.

Total retransmissions

$$= 8 \times 1$$

$$= \mathbf{8 \text{ frames}}$$

Answer

- Number of windows = **8**
- Total retransmissions = **8 frames**

Explanation

Flow control synchronizes the transmission speed between the sender and receiver. It prevents congestion, reduces frame loss, improves bandwidth utilization, and ensures efficient communication using the sliding window protocol.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given:

- Video size = 600 MB
- Synchronization checkpoint every 60 MB
- Failure occurs after 390 MB

Calculation

Step 1: Calculate completed checkpoints

$$390 \div 60$$

$$= 6.5$$

Completed checkpoints

$$= 6$$

Step 2: Last successful checkpoint

$$6 \times 60$$

$$= 360 \text{ MB}$$

Step 3: Data to be retransmitted

390 – 360

= **30 MB**

Answer

- Number of checkpoints created = **6**
- Data retransmitted after recovery = **30 MB**

Explanation

The **Session Layer** establishes, manages, and terminates communication sessions. Synchronization checkpoints divide long transmissions into recovery points. When a failure occurs, transmission resumes from the last checkpoint instead of restarting from the beginning. This reduces retransmission time, saves bandwidth, and improves communication reliability.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original file size = **900 MB**
- Compression = **40%**
- Encryption overhead = **5%** of the compressed file size

Calculation

Step 1: Calculate the compressed file size

Compressed file size

$$= \text{Original file size} \times (100 - \text{Compression}) \div 100$$

$$= 900 \times 60 \div 100$$

$$= \mathbf{540 \text{ MB}}$$

Step 2: Calculate the encryption overhead

Encryption overhead

$$= 540 \times 5 \div 100$$

$$= \mathbf{27 \text{ MB}}$$

Step 3: Calculate the encrypted file size

Encrypted file size

$$= \text{Compressed file size} + \text{Encryption overhead}$$

$$= 540 + 27$$

$$= \mathbf{567 \text{ MB}}$$

Step 4: Calculate the total percentage reduction

Percentage reduction

$$= ((\text{Original file size} - \text{Encrypted file size}) \div \text{Original file size}) \times 100$$

$$= ((900 - 567) \div 900) \times 100$$

$$= (333 \div 900) \times 100$$

$$= 37\%$$

Answer

- Compressed file size = **540 MB**
- Encrypted file size = **567 MB**
- Total percentage reduction = **37%**

Explanation

The **Presentation Layer** is responsible for data compression, encryption, and format conversion. Compression reduces the amount of data transmitted, improving bandwidth utilization. Encryption protects data from unauthorized access during transmission. Thus, the Presentation Layer enhances both communication efficiency and security.

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given:

- Data transferred = **3 GB**
- Effective throughput = **120 Mbps**
- Request size = **3 MB**

Calculation

Step 1: Convert GB into MB

$$3 \text{ GB} = 3000 \text{ MB}$$

Step 2: Calculate the number of requests

Number of requests

$$= \text{Total data} \div \text{Request size}$$

$$= 3000 \div 3$$

$$= 1000 \text{ requests}$$

Step 3: Calculate the transmission time

Convert data into Megabits

$$3000 \text{ MB} \times 8$$

$$= 24,000 \text{ Megabits}$$

Transmission time

$$= \text{Total Megabits} \div \text{Throughput}$$

$$= 24,000 \div 120$$

$$= 200 \text{ seconds}$$

Answer

- Number of requests = **1000**

- Transmission time = **200 seconds**

Explanation

The **Application Layer** provides network services directly to end users. FTP enables reliable file transfer by managing file requests, user authentication, and communication between the client and server. It ensures accurate, secure, and efficient file transfer over the network.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given:

- Application Layer delay = **4 ms**
- Transport Layer delay = **3 ms**
- Network Layer delay = **5 ms**
- Data Link Layer delay = **2 ms**
- Physical Layer delay = **1 ms**
- Propagation delay = **18 ms**
- Transmission delay = **12 ms**

Calculation

Step 1: Calculate the total processing delay

Processing delay

$$= 4 + 3 + 5 + 2 + 1$$

$$= \mathbf{15\ ms}$$

Step 2: Calculate the total end-to-end delay

Total delay

$$= \text{Processing delay} + \text{Propagation delay} + \text{Transmission delay}$$

$$= 15 + 18 + 12$$

$$= \mathbf{45\ ms}$$

Answer

- Processing delay = **15 ms**
- Total end-to-end delay = **45 ms**

Explanation

Each OSI layer contributes to successful communication. The Application Layer provides user services, the Transport Layer ensures reliable delivery, the Network Layer performs routing, the Data Link Layer manages framing and error detection, and the Physical Layer transmits bits through the communication medium. Together, these layers determine the total communication delay.

9.A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total

number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

- Useful data = **80 MB = 80,000,000 bytes**
- Frame size = **1600 bytes**
- Protocol overhead = **80 bytes per frame**

Calculation

Step 1: Calculate the useful data per frame

Useful data per frame

= Frame size – Overhead

= 1600 – 80

= **1520 bytes**

Step 2: Calculate the number of frames

Number of frames

= Total useful data ÷ Useful data per frame

= 80,000,000 ÷ 1520

= 52,631.58

≈ **52,632 frames**

Step 3: Calculate the total protocol overhead

Total overhead

$$= \text{Number of frames} \times \text{Overhead per frame}$$

$$= 52,632 \times 80$$

$$= \mathbf{4,210,560 \text{ bytes}}$$

Step 4: Calculate the transmission efficiency

Transmission efficiency

$$= (\text{Useful data per frame} \div \text{Frame size}) \times 100$$

$$= (1520 \div 1600) \times 100$$

$$= 0.95 \times 100$$

$$= \mathbf{95\%}$$

Answer

- Number of frames = **52,632**
- Total protocol overhead = **4,210,560 bytes**
- Transmission efficiency = **95%**

Explanation

Protocol overhead consists of headers and trailers added by different OSI layers. Although overhead reduces the effective data rate slightly, it is necessary for addressing, routing, error

detection, and reliable communication. Efficient protocol design minimizes overhead while maintaining network performance and reliability.

10.A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given:

- Original medical record size = **240 MB**
- Compression = **30%**
- Segment size = **1400 bytes**
- Data Link Layer overhead = **60 bytes per frame**

Calculation

Step 1: Calculate the compressed file size

Compressed file size

$$= \text{Original size} \times (100 - \text{Compression}) \div 100$$

$$= 240 \times 70 \div 100$$

$$= \mathbf{168 \text{ MB}}$$

Step 2: Convert compressed file into bytes

168 MB

= **168,000,000 bytes**

Step 3: Calculate the number of segments

Number of segments

= $168,000,000 \div 1400$

= **120,000 segments**

Step 4: Calculate the total protocol overhead

Protocol overhead

= Number of segments $\times$ Overhead per frame

= $120,000 \times 60$

= **7,200,000 bytes**

= **7.2 MB**

Step 5: Calculate the final amount of data transmitted

Final transmitted data

= Compressed file size + Protocol overhead

= $168 + 7.2$

= **175.2 MB**

Answer

- Compressed file size = **168 MB**
- Number of segments = **120,000**
- Total protocol overhead = **7.2 MB**
- Final data transmitted = **175.2 MB**

Explanation

The **Presentation Layer** compresses the medical record to reduce its size and may encrypt it to ensure confidentiality. The **Transport Layer** divides the compressed data into segments for reliable transmission. The **Data Link Layer** encapsulates each segment into frames by adding headers and trailers for error detection. Finally, the **Physical Layer** transmits the data through the communication medium. Together, these layers provide secure, reliable, and efficient communication in an IoT-enabled healthcare network.